

Systems Design

Document role

- This file owns the active subsystem baseline and cross-system modeling context for the build.
- Use this file to describe what the current system is, how the major subsystems fit together, and which assumptions are still active.
- Keep the final 48V house/alternator architecture, shutdown order, and control-path lock in docs/core/ELECTRICAL_48V_ARCHITECTURE.md .
- Keep exact conductor, fuse, layout, and install-level detail in docs/implementation/ .
- Keep decision/risk/open-question state in docs/core/TRACKING.md rather than letting this file become the live issue tracker.

Electrical and energy

Goals

- Deliver workday reliability with reserve margin
- Keep wiring safe, labeled, and serviceable

Electrical doc links

- Canonical 48V architecture and alternator-control baseline: [ELECTRICAL 48V ARCHITECTURE](#)
- Implementation topology (components, fuses, holders, gauges): [ELECTRICAL overview diagram](#)
- Fuse IDs, locations, housing methods, spares, and BOM mapping: [ELECTRICAL fuse schedule](#)
- Voltage architecture trade study (12V vs 48V): [ELECTRICAL 12V vs 48V trade study](#)
- Alternator architecture trade study history (research archive only; final decisions moved to the canonical 48V architecture doc): [ELECTRICAL 48V dual alternator trade study](#)
- Solar option screening matrix (stringing + MPPT fit flags): [SOLAR configuration matrix](#)
- Electrical decisions, risks, and unresolved items: [TRACKING](#)

Planning snapshot (as-of 2026-05-14)

- Battery bank: 3x 48V 100Ah LiFePO4 from BOM row 3 (15.36 kWh nominal at 51.2V battery nominal).
- House architecture: 48V core with Orion-Tr Smart 48V->12V charging/step-down feeding a shared battery-backed 12V junction.
- Inverter/charger candidate: Victron MultiPlus-II 48/3000/35-50 .
- Charge sources in current BOM: solar MPPT, dedicated 48V secondary alternator path (Mechman + WS500 + APM-48 migration baseline), shore AC charger path.
- Monitoring and protection: Cerbo GX, SmartShunt, battery temp sensing, Class T primary fuse + branch fusing.
- AC protection chain is now purchased/locked for Phase 1 (shore source/adapters -> portable EMS -> shore cord -> L5-30 inlet -> single 6-way AC DIN enclosure -> 30A AC-in breaker -> MultiPlus -> 30A AC-out main -> two 20A GFCI branches), with AC-in-only initial charging still used as the first live validation step.
- Current build phase: use the installed camper shell for dead-mechanical electrical layout, measured block envelopes, and service-map validation before final cable cuts, energization, extrusion vendor cuts, skins, or finish-floor glue-down.

Modeling rules (procurement-first plus full-load)

- Primary procurement source of truth is bom/bom_estimated_items.csv .
- Load model is maintained in bom/load_model_wh.csv (model v3) and includes BOM-sourced installed loads plus owner-supplied work electronics (kept out of BOM cost totals).
- Legacy workbook WH model assumptions are retired and not used.
- Voltage convention: use 48V as architecture label, but use 51.2V nominal for battery Wh accounting.
- Run-length convention: measured physical layout lengths are cut-length source-of-truth; CAD values are planning references only.

Input reference (maintained)

Input	Current value	Source
Battery bank	3x Dufume 51.2V 100Ah (1S3P; manual allows up to 1S4P)	bom/bom_estimated_items.csv row 3 + references/Dunfume_36V_48V_100Ah_Battery_-_User_Manual.pdf
Inverter/charger	MultiPlus-II 48/3000/35-50	bom/bom_estimated_items.csv row 12
Alternator charging	Dedicated 48v secondary alternator path (Mechman + WS500 + APM-48) with Upfitter #3 -> WS500 brown ignition manual control and Lynx Slot 3 alternator branch fuse lock	bom/bom_estimated_items.csv ROWS 168-171, 176 + docs/core/ELECTRICAL_48V_ARCHITECTURE.md

Sterling legacy items	BB1248120 + BBR retained physically only until Mechman confirmation, then returned	bom/bom_estimated_items.csv rows 18 and 26
Legacy single-12V upgrade path	Mechman 370A + Big 3 path is deprecated under the dual-48v migration baseline	bom/bom_estimated_items.csv rows 103 and 104
DC-DC charger	Orion-Tr Smart 48/12 30A (360W)	bom/bom_estimated_items.csv row 20
12V buffer battery	12V 100Ah LiFeP04 on shared 12V junction (F-11 + SW-12V-BATT)	bom/bom_estimated_items.csv rows 21, 124, and 125
Solar array candidate	Flexible-first placeholder (~800-1000W class; prior 9x100W/3S3P concept retained for modeling basis)	bom/bom_estimated_items.csv row 24
Solar controller	SmartSolar MPPT 150/45	bom/bom_estimated_items.csv row 25
Load profiles (BOM + owner-supplied office loads)	core_workday, winter_workday, minimal_idle_day	bom/load_model_wh.csv
Owner-supplied office assumptions	Laptop + 27 inch 1440p monitor + tablet/peripheral charging	bom/load_model_wh.csv rows marked Owner-Supplied

Modeled expected usage

Load totals below are from `bom/load_model_wh.csv` model v3 (BOM loads plus owner-supplied office loads).

Scenario	Daily energy	5-day workweek energy	7-day week energy	Dominant contributors
core_workday	3,530 Wh	17,650 Wh	24,710 Wh	Laptop, monitor, Starlink, cooking, inverter idle
winter_workday	4,202 Wh	21,010 Wh	29,414 Wh	Laptop, monitor, Starlink, diesel heater, cooking
minimal_idle_day	624 Wh	3,120 Wh	4,368 Wh	Fridge + always-on monitoring/detector loads

Capacity analysis (corrected battery bank)

Metric	Formula	Result
Nominal battery energy	51.2V x 100Ah x 3	15,360 Wh
Usable energy to 20% reserve floor	15,360 x 0.8	12,288 Wh
Core day depth of discharge	3,530 / 15,360	22.98% per day
Winter day depth of discharge	4,202 / 15,360	27.36% per day

Autonomy (no charging)

Scenario	Days (100% -> 20%)
core_workday	3.48
winter_workday	2.92
minimal_idle_day	19.69

Charging potential

All values are planning-level and should be replaced with measured charge logs after shakedown tests.

Solar charging (900W flexible 3S3P focus)

Base planning factor for your target roof setup is now `68%` end-to-end harvest efficiency, with sensitivity from `60%` (poor conditions) to `75%` (strong conditions).

Derate component	Planning factor	Note
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Nameplate realization	0.95	Manufacturing variance and real-world rating spread
Thermal factor (flexible with air gap)	0.90	Better than bonded-flat flexible, still hotter than rigid stand-off
MPPT + wiring efficiency	0.96	Controller and cable losses
Flat-roof angle/azimuth factor	0.88	No seasonal tilt optimization
Soiling + mismatch + partial shading allowance	0.93	Dirt, string mismatch, and intermittent shade impacts
Combined planning factor	~0.68	Product of factors above

900W array efficiency	2 PSH day	4 PSH day	5 PSH day	Net vs core_workday at 4 PSH	Net vs winter_workday at 4 PSH
60%	1,080 Wh	2,160 Wh	2,700 Wh	-1,370 Wh/day	-2,042 Wh/day
68% (planning base)	1,224 Wh	2,448 Wh	3,060 Wh	-1,082 Wh/day	-1,754 Wh/day
75%	1,350 Wh	2,700 Wh	3,375 Wh	-830 Wh/day	-1,502 Wh/day

- Break-even PSH at 900W :
- core_workday (base 68%): 3,530 / (900 x 0.68) = 5.77 PSH/day.
- winter_workday (base 68%): 4,202 / (900 x 0.68) = 6.87 PSH/day.
- Sensitivity band for winter_workday : 6.23 PSH/day (75%) to 7.78 PSH/day (60%).
- MPPT 150/45 is not the bottleneck for the current array range.

Alternator charging (dedicated 48V secondary alternator path)

- Active migration baseline: Mechman dual-alternator kit + WS500 regulator + APM-48 protection module.
- Lynx Slot 3 alternator branch fuse (F-04) is now locked at 150A (58V/80V MEGA class).
- WS500 low-current fuse set is now explicitly part of the design baseline (10A power lead baseline, 3A sense, 5A current-sense where required).
- Manual charge-enable/disable path is now locked to Ford Upfitter Switch #3 feeding the WS500 brown ignition/enable wire through local inline fuse F-15 .
- WS500 white Feature-In is reserved for future automatic fault-interlock work and is not required in Phase 1.
- Cable decision lock for this pass: reuse existing uncut 2/0 inventory for the alternator charge path (~20 ft one-way assumed), with dedicated equal-size negative run.
- Sterling BB1248120 / BBR remain physically on hand only as return-pending hardware until Mechman fitment/content confirmation closes.
- Open execution gate: contact Mechman first, then complete Sterling physical return workflow.

Shore charging (MultiPlus-II charger path)

- Charger limit from model string: 35A .
- Existing planning math uses 56.8V absorption target (~1,988W), while the battery manual/BOM basis references 58.4V (~2,044W at 35A). Reconcile and document the actual MultiPlus LiFePO4 charge profile before first energization.
- Initial garage workflow: use AC-in-only MultiPlus charging and connect/charge one 48V battery at a time before paralleling the 1S3P bank.
- Ideal bulk-only recharge times at 56.8V planning basis:
- Replace one core_workday : 1.78h .
- Replace one winter_workday : 2.11h .
- Recharge full 3x bank from 20% to 100% : 6.18h .
- Recharge one 48V 100Ah battery from 20% to 100% : about 2.06h ideal bulk-only.
- Real-world times are longer due to absorption taper near full charge, lower input-current limits on 15A household sources, and any configured charge-current derate.

Operational implications and constraints

- Battery capacity now supports roughly 2.9-3.5 office-workdays without charging depending on season and reserve policy.
- With 900W flexible solar at the base 68% factor, 4 PSH leaves a material daily deficit for both core_workday and winter_workday .
- Shore charging can materially recover SOC in a single evening (~6.18h from 20% to 100% in bulk-ideal terms).
- Alternator recovery potential is expected to materially exceed the old Sterling ~1.5kW ceiling once the dedicated 48V alternator path is commissioned.
- Current execution risk is no longer charger-capacity-limited operation; it is migration/commissioning quality (fitment, regulation, protection, and measured thermal behavior).
- MultiPlus-II 48/3000 inverter continuous output (~2,400W) can be exceeded by simultaneous induction + microwave + other AC loads, so high-draw AC loads need sequencing.

- Orion-Tr Smart 48->12V 30A charger (360W) is the continuous charging/feed ceiling into the shared 12V junction; buffer battery supports short transients but sustained overload still requires load budgeting.

Safety baseline

- Positive path sequence: battery -> Class T fuse (near source) -> disconnect -> Lynx Distributor -> fused branch feeds
- Negative path sequence: battery -> SmartShunt -> Lynx Distributor negative bus -> all load returns on load side of shunt
- Dedicated alternator branch grounding rule set: run equal-or-larger dedicated negative cable from secondary alternator to house-bank return path and avoid sheet-metal return paths.
- If the truck uses an RVC ground-sensor loop, route the upgraded ground path through the loop per vehicle requirements.
- Battery thermal strategy: insulated battery box, ducted warm-air branch, thermostat/relay enable logic
- Wiring practices: grommets, loom, glands, abrasion protection, and bend-radius validation
- Reference links from workbook notes:
 - https://youtu.be/dSYKabw_rgs?t=651
 - <https://www.diodeled.com/45-channel.html>

Electrical overview diagram (implementation)

- Full implementation-level topology diagram: `docs/implementation/ELECTRICAL_overview_diagram.md`
- Scope includes fuse IDs, fuse-holder/housing methods, branch wire-gauge selections, and documented sizing assumptions.

RF-003 Distribution Topology Decision (Lynx Locked)

Decision date: 2026-02-12

Approved architecture for Phase 1:

- Victron Lynx Distributor M10 (LYN060102010) is the single distribution backbone.
- Current tracked unit price is \$192.47 .
- Current modeled 48V fused branches (4 total):
- MultiPlus-II 48/3000
- SmartSolar 150/45
- Dedicated 48V alternator branch output (F-04 locked 150A)
- Orion-Tr Smart 48/12 (shared 12V junction feeder)
- 1x Lynx Distributor covers current branch count with 0 spare fused outputs.
- If future branch expansion is needed, add a second Lynx module in that phase.

Implementation notes:

- Lynx Distributor includes the negative busbar, so a separate standalone negative bus is not required in the Lynx path.
- Lynx Distributor branch-fuse LEDs need a Lynx Shunt VE.Can or Lynx Smart BMS ; with SmartShunt -only monitoring, LED fuse indication is not active.
- Main Class T protection baseline is 3x (one per battery-positive conductor) and is tracked in the fuse schedule/BOM.

Reference links:

- Lynx Distributor manual/specs: https://www.victronenergy.com/media/pg/Lynx_Distributor/en/introduction.html
- Lynx Distributor retail example: <https://www.invertersrus.com/product/victron-lynx-distributor/>

Fuse Determination Baseline (Lynx)

Objective: maintain a complete start-to-finish fuse schedule for the approved Lynx architecture.

Current detailed schedule (active reference):

- `docs/implementation/ELECTRICAL_fuse_schedule.md`

Scope covered:

1. Main battery protection (Class T) quantity, rating, and placement.
2. Lynx branch fuses for each 48V branch (MultiPlus, MPPT, dedicated alternator branch, Orion input).
3. WS500 low-current fuse requirements and alternator-branch protection coordination.
4. Any additional protective devices required by manufacturer manuals for both charge-source and load paths.

Method:

1. Pull max current requirements and overcurrent guidance from each device manual.
2. Confirm planned wire gauge/length/insulation assumptions for each run.

- 3. Size each fuse to protect the conductor first while meeting equipment requirements.
- 4. Build final fuse matrix with part numbers, quantities, and BOM row mapping.

Solar

- Current direction: flexible-first design path due roof 75 lb hard panel-weight cap.
- Near-term project policy: keep solar final procurement/string lock deferred deeper into build while preserving routing/passthrough reservations now.
- Hardwall popup wiring baseline for planning: no hidden in-wall solar run; use exterior-rated coiled jumper cable(s) from roof solar exit to a lower-shell weatherproof passthrough.
- BOM scope for that routing method is tracked in `bom/bom_estimated_items.csv` row 121 .
- Open points: exact passthrough location, connector standard (MC4 direct vs bulkhead adapter strategy), and final flexible module/string configuration before SKU lock.
- Energy takeaway from current modeled load: moving from sub- 600W class toward ~800-1000W flexible can materially reduce daily deficit, but modeled office-workday profiles still require charging strategy integration (solar + alternator + shore).

Electrical reference maintenance workflow

- Update trigger conditions:
 - Any change to battery, inverter/charger, DC-DC, or solar components in `bom/bom_estimated_items.csv` .
 - Any measured field data that materially changes duty-cycle assumptions.
 - Any change in appliance selection, owner-supplied office electronics, or expected duty cycle in `bom/load_model_wh.csv` .
- Update process:
 1. Update component rows and assumptions in `bom/load_model_wh.csv` .
 2. Recalculate scenario daily Wh totals in this file from that CSV.
 3. Recalculate autonomy at the active reserve floor (usable Wh / daily Wh).
 4. Recalculate charging tables with latest charge source ratings and assumptions.
 5. Record what changed in `docs/core/TRACKING.md` (decision/risk/open questions) and `logs/LOG.md` .
 6. Remove stale assumptions that are not backed by BOM rows or explicit owner-supplied load assumptions.
- Formula quick reference:

```
battery_nominal_wh = battery_voltage_v * battery_capacity_ah * battery_quantity
daily_wh = sum(component_wh_per_day) + conversion_loss_wh
autonomy_days = usable_battery_wh / daily_wh
solar_daily_wh = array_watts * effective_psh * solar_efficiency_factor
solar_efficiency_factor = nameplate_realization * thermal_factor * mppt_wiring_factor * orientation_factor * soiling_shading_factor
alternator_daily_wh = dc_dc_output_watts * drive_hours
alternator_practical_w = (alternator_rated_a - vehicle_base_load_a) * alternator_voltage_v * conversion_efficiency
house_charge_current_a = alternator_practical_w / charge_voltage_v
shore_charge_power_w = charge_voltage * charger_current_a
bulk_charge_hours = energy_to_replace_wh / shore_charge_power_w
```

- Retired model note:
- `bom/load_model_wh.csv` v1 (workbook-derived chart) and v2 (BOM-only) are superseded by model v3 (BOM plus owner-supplied office loads).

Communications

- Primary internet path: Starlink Standard (DC power conversion item tracked in BOM)
- Secondary/fallback path: TBD (cellular path and carrier diversity not frozen)
- Placement constraint: routing and passthrough location unresolved

Plumbing (if included in phase 1)

- Near-term baseline: decouple cold-water galley build from final hot-water selection. Build around tank, pump, faucet/sink, drain/graywater path, service shutoffs, and a capped future hot-water tie-in.
- Interior layout and wet-spine draft owner: [INTERIOR furniture layout and galley](#). Current direction is cold-water-first with a passenger-side lofted fridge/wet-spine exoskeleton: Iccco/fridge raised about 16 in , pump/accumulator/service board below the fridge next to the wheel-well tank, rear shower/fill/vent service access, graywater cassette, and capped future hot-water stubs.
- Water capacity: purchased 36 gal wheel-well tank. Full water mass is about 300 lb before tank/brackets/plumbing; plan around roughly 320+ lb installed when full.
- Tank restraint posture: the wheel-well form factor is materially better than a tall/skinny vertical tank because it lowers center of gravity and reduces overturning moment, but plusnut wall brackets should be verified as part of the restraint system rather than assumed as the only crash/off-road load path.

- Pump candidate captured: Shurflo 3.0 GPM class.
- Faucet is now an explicit missing purchase class and should be tracked separately from generic cabinetry.
- Hot-water decision posture: electric tankless is out of scale for the current 48/3000 inverter; small tanked electric is plausible later but adds AC load-management complexity; portable propane tankless remains provisional outdoor-use only; listed indoor/RV propane is deferred until venting/combustion-air/clearance package is locked.
- Freeze and winterization strategy: draft around visible service controls, low-point drains, blowout port, winterization pickup, capped future heater bypass/stubs, no hidden low loops, and rear/aisle service access; final details remain measurement-gated.
- Gravity-fill vent hose correction: owner-measured vent nipple is about 10 mm OD on the main land and 11 mm OD at the largest barb/ridge. Specify 10 mm ID food-grade/potable tube, or 3/8 in ID as the common inch fallback; the previously ordered 1/2 in ID x 5/8 in OD tube is oversized.

Cabinetry and structure

- Interior furniture/layout draft owner: [INTERIOR furniture layout and galley](#). Recommended planning direction is the office-first hybrid: passenger-side lofted Iceco/fridge on an extrusion exoskeleton, pump/accumulator below the fridge next to the 36 gal wheel-well tank, adjacent separated battery bench, driver-side electrical closet/workstation/DC shelf, diesel heater low in the driver-side utility zone, and clear center aisle/cabover movement.
- Driver-side workstation implementation draft: [INTERIOR driver side workstation](#). It captures the current best direction for a driver-side service-spine desk, stow-low monitor cassette/rising VESA spine, electrical-closet/DC-shelf interface, shallow storage, travel locks, cable management, heat/noise control, and roof-close acceptance tests.
- Aluminum extrusion strategy: the broad 15-series starter order is superseded. Default to 10-series prototype stock/hardware for furniture where practical; reserve larger extrusion for measured heavy/dynamic freestanding modules that prove they need the stiffness, such as the lofted fridge skeleton, electrical closet frame, monitor mast/spine, or battery bench structure if dry-fit proves the load path needs it.
- Panel strategy: living-facing furniture surfaces should use mechanically removable overlay panels over the frame; service zones stay exposed or quick-removable. Magnetic service covers are acceptable only for light non-structural panels and need locator/anti-shear features plus backup retention where loss would matter.
- Current furniture CAD and May 4 generated diagrams are reference-only after installed-shell layout changes; re-CAD block envelopes around the passenger-side lofted fridge/wet-spine, tank overlap, battery bench, driver-side electrical closet, and desk before final cut lists.
- Modular mounting baseline now includes T-slot/strut rails both exterior (recovery/tool mounts like shovel/Maxtrax) and interior (baskets/hooks/tie-down points); BOM rows 119 and 120 .
- Drawer hardware baseline includes 4x soft-close undermount slide kits for primary cabinetry drawers; BOM row 122 , but final lengths are deferred until fridge/tank/electrical module envelopes are verified.
- Desk concepts captured: fixed full-time work surface with possible auxiliary fold leaf; Lagun-style or pneumatic concepts should remain secondary unless the measured work envelope proves they are stable enough.
- Material ideas captured: phenolic/richlite top, sound treatment, panel anti-rattle tape, frosted/angled LED strips, smoke-grey acrylic shallow cubby covers.
- Monitor travel strategy concept: stow-low/face-down assisted deployment with structural bracing, hard stops, positive travel locks, cable drag chain, and a visible roof-safe state before pop-down.

HVAC and condensation

- Heating approach in notes: diesel heater low in the driver-side utility/electrical-closet zone, with any warm-air branch to batteries controlled and kept off sensitive electrical hardware.
- Diesel fuel planning: owner is investigating an external truck-bed-wall tank around 8 gal in an approximately 3 in x 17 in envelope. Preferred direction is to keep diesel storage/fill/pump outside the living space, with only a protected fuel line passing through a grommet/bulkhead to the heater. 8 gal diesel reserve is roughly 50-60 hr of continuous full-output runtime for a typical 5 kW heater; use ~48 hr as a conservative full-blast planning number.
- Ventilation: Maxxair fan included in current camper config
- Lighting split: Hiatus factory overhead LED+dimmer is a separate circuit from planned ambient/cabinet LED strips (Govee)
- Condensation controls and climate envelope limits: TBD

Safety

- Purpose: define a practical, build-ready safety baseline for the current architecture (48V 15.36kWh house bank, 12V distribution, 120VAC shore/inverter path, and propane-supported heating/hot-water concepts).
- Priority order: prevent ignition and overcurrent faults, preserve safe shutdown paths, detect hazards early, and make isolation/service repeatable.
- Final install gate: before energizing or using propane in service, verify all items against manufacturer manuals and complete licensed inspection where required.

System-wide controls

- Keep one-line diagrams, fuse IDs, and conductor IDs synchronized across:
 - docs/implementation/ELECTRICAL_overview_diagram.md
 - docs/implementation/ELECTRICAL_fuse_schedule.md
 - docs/core/TRACKING.md
- Ensure all protection and isolation devices are physically accessible without disassembling fixed furniture.
- Keep gas components and AC/DC electrical components separated by design; no mixed service cavities without physical barriers and clear labeling.

- Label every branch and shutoff point so an operator can isolate faults quickly under stress.

48V battery system safety (primary)

- Main hazards: high fault current, sustained DC arc potential, short-circuit heating, incorrect polarity during service, and thermal stress events.
- Required architecture controls:
- Battery positive path stays: battery -> Class T fuse near source -> main disconnect -> Lynx fused branches.
- Battery negative path stays: battery -> SmartShunt -> Lynx negative bus (all returns on load side of shunt).
- Use only voltage-appropriate overcurrent devices on house DC branches (58V / 80V class on 48V paths); do not substitute 32V automotive-only fuses on 48V circuits.
- Keep all busbars/studs covered and insulated; use boot covers, strain relief, and abrasion protection on all near-bus runs.
- Manual alternator shutdown order stays: Upfitter #3 OFF first to disable the WS500 , then open the main 48V disconnect only after alternator charging is no longer active.
- Commissioning controls (first energization and after major rework):
 1. Verify polarity and expected voltage at each segment before inserting branch fuses.
 2. Confirm torque marks on all high-current terminals and re-check after initial thermal cycles.
 3. Confirm the disconnect fully de-energizes downstream service zones as intended.
 4. Validate no unintended parallel return paths bypassing shunt measurement.
- Service controls:
- Remove conductive jewelry, use insulated tools, and keep one-hand work practice on live-exposure checks.
- Never perform branch rewiring with battery disconnect closed unless the specific test requires energized state and a spotter is present.

12V distribution safety

- Main hazards: feeder overload from Orion output limits, hidden voltage drop causing heat at terminations, and unfused accessory additions.
- Required controls:
- Keep Orion output feeder fused at source (F-07) and avoid adding unfused taps between Orion and the shared 12V junction.
- Keep 12V buffer battery positive protected at source (F-11) and route service isolation through SW-12V-BATT downstream of F-11 .
- Keep SW-12V-BATT in its normal closed position during operation; use open position only for service isolation/diagnostics.
- In normal closed operation, Orion supports both active 12V loads and buffer-battery maintenance through the shared junction path.
- In this baseline, the 12V fuse block is the shared junction device (main + stud combine point plus integrated negative bus return point).
- Do not solder-splice high-current 12V source conductors; use crimped lugs on rated stud terminals.
- Maintain branch-level fuse-to-conductor coordination per docs/implementation/ELECTRICAL_fuse_schedule.md .
- Keep always-on detector branch (12V-05) protected but never switch-controlled.
- Keep ambient/cabinet strip lighting on the dedicated DC branch (12V-11) so low-light use does not require inverter operation.
- If sustained 12V demand exceeds Orion headroom, treat additional 48V->12V charger capacity (BOM row 118) as a safety action, not a convenience upgrade.

120VAC shore/inverter safety

- Main hazards: shock from miswired neutral/ground, ground-fault exposure at outlets, and overcurrent heating from undersized branch protection.
- Required controls:
- Keep shore path order: shore source/adapters -> portable EMS/surge protection -> shore cord -> shore inlet (L5-30) -> combined AC DIN enclosure (30A UL489 AC-in breaker/disconnect) -> MultiPlus AC-in.
- Keep AC-out path order: MultiPlus AC-out-1 -> 10/3 feeder -> combined AC DIN enclosure (30A AC-out main + 20A / 20A branch breakers) -> GFCI receptacle on each active branch.
- Keep AC-in hardware on a 30A / 10 AWG basis, and set MultiPlus input current limit to the actual source (15A , 20A , or 30A) whenever adapters are used.
- Keep AC-out-2 as reserve-only in Phase 1 (labeled capped route; no energized branch hardware yet).
- Preserve continuous equipment grounding and chassis bond through all AC paths.
- Do not add a fixed downstream neutral-ground bond in branch wiring; neutral/ground behavior must follow MultiPlus transfer/bonding design.
- Commissioning checks:
 1. AC-in-only charger validation with AC-out loads disconnected for first battery charge.
 2. Receptacle polarity test on each outlet location.
 3. GFCI/RCD trip test at each protected branch.
 4. Verify AC-out-2 remains de-energized as a reserve-only capped route in Phase 1.

Propane safety (rear-mounted tank plus passthrough and hot-water path)

- Main hazards: leak accumulation, ignition near electrical equipment, CO exposure, and using outdoor-only appliances in enclosed space.
- Rear-mount tank controls:
- Keep cylinder upright and externally mounted with impact-resistant bracketry and valve protection.

- Keep tank shutoff valve accessible from outside without tools.
- Protect hose/regulator routing from road debris, abrasion, and exhaust heat zones.
- Propane passthrough controls:
- Use sealed bulkhead/pass-through hardware with chafe protection at all penetrations.
- Avoid concealed unions/joints in inaccessible cavities; keep serviceable connections at inspection points.
- Perform leak tests after any connection change and before each trip phase where propane is used.
- Appliance selection rule (critical):
- Treat portable outdoor tankless heaters as outdoor-use-only unless a specific model is explicitly listed for indoor/RV enclosed installation with compliant venting.
- If an indoor propane water-heating path is pursued, lock to an RV/marine-listed indoor unit with approved venting/combustion-air method and documented install clearances before purchase.
- Detection and ventilation controls:
- Keep an LP detector low in cabin, CO detector in breathing zone, and smoke detector high in cabin.
- Test detector alarm functions on a recurring schedule and replace by manufacturer expiration date.

Emergency shutdown baseline

1. Remove active high-draw loads (AC and gas appliances) if safe to do so.
2. Open the 48V main disconnect and remove shore input.
3. Close propane cylinder valve at the tank.
4. Ventilate cabin area and confirm detectors are active.
5. Use fire extinguisher only if contained/incipient and exit path is clear; otherwise evacuate and call emergency services.
6. Do not re-energize or reopen gas supply until fault root cause is identified and corrected.

Safety hold points before walls/panels are closed

- Complete and document high-current DC inspection (polarity, fuse value, terminal torque, insulation/boots).
- Complete AC verification (polarity, GFCI/RCD trip tests, branch labeling, ground continuity).
- Complete propane leak check and pressure-hold verification with all planned valves/fittings in final positions.
- Complete detector placement and functional alarm tests.
- Record evidence in logs/LOG.md and track unresolved items in docs/core/TRACKING.md .

Source artifacts

- docs/legacy/SYSTEMS_workbook_build_notes_obsolete.md
- bom/load_model_wh.csv